

Editorial: Problem G - The Discipline Committee's Struggle

Setter: Hudson

Problem Overview

We need to support Range Add, Range Min (ChMin), Range Sum, and **Range Historic Sum** on an array.

Difficulty Analysis

This problem combines two very advanced Segment Tree techniques:

1. **Segment Tree Beats:** Required to handle Range ChMin updates efficiently.
2. **Historic Information Maintenance:** Required to track the sum of values over time.

Because the "ChMin" operation selectively updates only values strictly greater than x , the history accumulation becomes conditional. This requires maintaining separate lazy tags for the maximum elements and the non-maximum elements.

Solution Approach

1. Segment Tree Beats Logic

We maintain $max1$ (maximum value), $max2$ (strict second maximum), and cnt (count of max values) in each node. For an update $A_i = \min(A_i, x)$:

- If $x \geq max1$, ignore.
- If $max2 < x < max1$, update $max1$ to x . This is effectively adding $(x - max1)$ to all maximum elements.
- If $x \leq max2$, recurse.

2. Adding History

We treat the operations as separate affine transformations for:

- Class A: The elements equal to $max1$.
- Class B: The elements strictly less than $max1$.

We maintain four lazy tags:

- *lazy_A*: Value added to maximums.
- *lazy_B*: Value added to non-maximums.
- *hist_A*: Historic value accumulated by maximums.
- *hist_B*: Historic value accumulated by non-maximums.

When pushing tags down to a child:

- If child's *max1* matches parent's old *max1*, it inherits Class A tags.
- Otherwise, it inherits Class B tags.

Implementation

The complexity is $O(Q \log^2 N)$ or $O(Q \log N)$ depending on the potential function analysis, but practically it is fast enough for $N = 2 \cdot 10^5$.